

deciCARE: A Task-Driven Interface Design for Personalized Patient Education

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Abstract

Patients with hematologic malignancies often struggle to participate in treatment decisions due to complex care pathways. Although personalized education supports shared decision-making (SDM), nurses face time constraints, fragmented data, and repetitive manual work. To address these, we present deciCARE, a task-driven web prototype that unifies key steps in the patient education workflow. This system enables nurses to access patient information, plan treatment schedules, and automatically generate personalized educational materials from a searchable database, all within a single interface. An AI editing feature is also included to adapt educational content. The system improves information accessibility, reduces interaction overhead, and streamlines nursing workflows.

CCS Concepts

• **Human-centered computing** → **Web-based interaction; User interface design.**

Keywords

Personalized education; Nursing workflow; Interface design

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1 Introduction

Engaging in treatment decisions is challenging for patients with hematologic malignancies due to disease heterogeneity, uncertain prognosis, and rapidly evolving therapies [3, 10]. The complexity creates an information gap between patients and healthcare providers and hinders the development of standardized patient decision aids [4, 8, 10]. In this context, shared decision-making (SDM), a collaborative process between patients and clinicians, is crucial for improving treatment satisfaction and reducing decisional conflict [3, 10]. Oncology nurses play a key role in decision-making, helping patients interpret information, clarify values, and compare treatment options [9]. However, heavy workloads and limited time constrain their ability to assess needs and deliver personalized education [5, 7]. This challenge motivates our research question: How might we support nurses' educational work in clinical practice?

Our preliminary study aimed to improve the efficiency of nurse-led education to support SDM. We introduce deciCARE¹, an interactive web prototype that integrates clinical data, identifies patient distress, and auto-generates personalized educational materials. The system streamlines workflows, enabling nurses to effectively address information asymmetry, a key barrier to SDM [4, 8].

2 Informal Needfinding Interviews

We conducted interviews with three hemato-oncology nurses at a university hospital to identify their needs when supporting patients in SDM. Nurses face a significant challenge, although information exchange between patients and healthcare providers is essential to SDM, many patients remain passive or underinformed,

¹deciCARE: decisionCARE

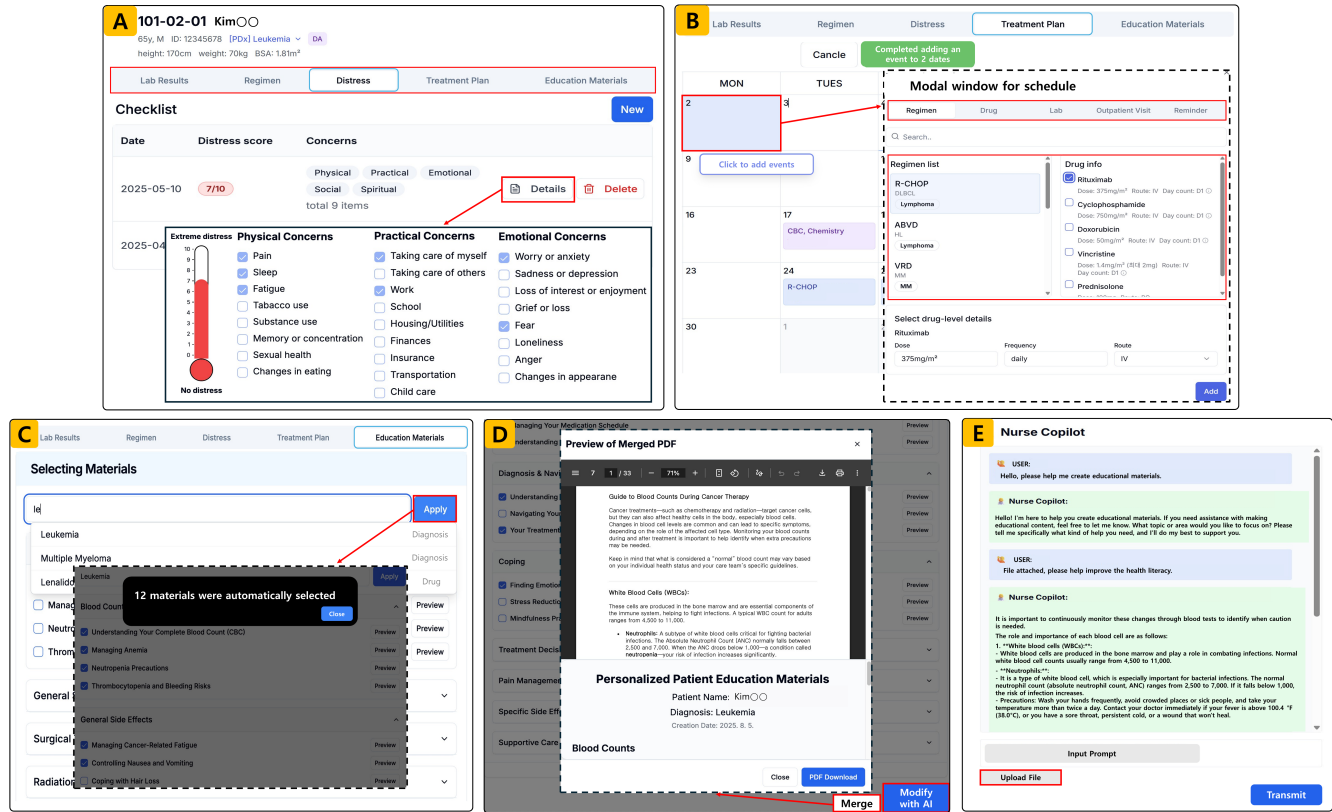


Figure 1: Task-based interaction flow in deciCARE. (A) Reviewing patient data in place (B) Scheduling treatments in bulk (C) Auto-selecting materials by keywords (D) Merging and downloading materials with one click (E) Modifying materials with AI

and providers often lack sufficient information to understand patients holistically. To address this challenge, we identified four key needs of nurses. First, nurses need to provide patients with reliable information that promotes patient engagement. Second, contextual factors such as social, psychological, and financial aspects are often implicit, making them difficult to access and integrate into care. Third, nurses expressed frustration with the repetitive manual work required to manage treatment schedules and customize educational materials, due to the complexity of chemotherapy. Lastly, while nurses recognize the importance of tailoring education to patients' treatment stage and health literacy level, limited time and staffing make such personalization difficult.

By analyzing the identified needs, we determined the design goals to be used during the prototype phase as follows: (1) presenting a consolidated view of clinical and contextual data; (2) streamlining generation of educational materials with minimal interaction; (3) supporting AI-assisted editing to adapt content to patients' contexts and literacy levels. The design goals are intended to reduce cognitive and operational load, enabling nurses to engage more effectively in personalized, patient-centered education.

3 Prototype Overview

deciCARE is a task-driven web interface designed to assist with hemato-oncology nurses in providing personalized patient education. It comprises four main tasks aligned with clinical workflows, each supported by integrated interface components (see Figure 1). **Task 1 (Figure 1A): Understanding patient contexts.** To support contextual understanding, deciCARE presents key patient information through a tab-bar interface, including diagnoses, laboratory results, prescribed regimens (e.g., drug names, dosages, and schedules), chemotherapy history, and distress scores. This tab-based navigation streamlines access to clinical and psychosocial data in a unified view, allowing nurses to efficiently review relevant information without switching screens or navigating across disparate systems.

Task 2 (Figure 1B): Planning the treatment schedules. The interactive calendar module enables nurses to intuitively register clinical events (e.g., chemotherapy sessions, medication administrations, diagnostic tests, outpatient visits, and reminders) through direct manipulation of the calendar interface, including clicking and dragging. Clicking a date on the calendar opens a modal window, where schedule details can be edited in place. The interface supports multi-date selection by dragging across dates and enables batch entry, reducing redundant operations. Its modular design minimizes interaction overhead and facilitates the efficient scheduling of treatment.

Task 3 (Figure 1C, D): Generating educational materials. deciCARE provides a keyword-based interface that matches materials to user input (e.g., diagnosis, regimen, drug name). Auto-complete suggestion supports accurate term entry, and a rule-based frontend structure selects relevant materials. These are presented as card components for preview and can be merged into a single PDF. This flow reduces manual effort and structures content generation, streamlining the creation of personalized education packets.

Task 4 (Figure 1E): Modifying materials with AI. To bridge the gap between clinical accuracy and patient comprehension, we developed an early version of Nurse Copilot, an AI-powered feature integrated into deciCARE. It leverages the GPT-4.0 API with prompts engineered to enhance patient understanding [1]. Nurse Copilot assists nurses in adapting content to patients' literacy levels. For example, it rephrases complex medical terminology into plain language, enhancing readability without compromising factual accuracy [1, 6].

4 Conclusion and Future Work

We developed deciCARE streamlining workflows, reducing interaction overhead, and supporting context-aware communication between the nurses and the patients. We outline two key directions for extension. First, we may extend Nurse Copilot beyond PDF rephrasing to support multimodal outputs, including text-to-speech and text-to-image generation for patients with disabilities (e.g., people with visual impairments, people with hearing loss). We also aim to advance its personalization capabilities to better tailor educational content to patients' health literacy. Second, we plan to expand deciCARE to conditions, such as mental health disorders and chronic illnesses (e.g., diabetes) where personalized education is necessary for supporting effective long-term self-management [2].

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